

Disclosure about the use of LLM's by Noortje van Maldegem

My teammates and I were very excited about this assignment, because we are really interested in Computational Neuroscience. Therefore, we went very deep in our analysis and in all our research to create a biologically plausible model. However, we unfortunately had only a couple of days to do our research and describe all our findings. I am not a fan of letting AI do all the work, because I really want to learn something from the assignments, and you do not learn if AI does everything. However, since we had some time pressure and we wanted to make our analysis and research as complete as possible, I used AI more than I usually want to do. Especially for writing the report I needed some assistance in how to formulate all the findings in a good way. Writing is really one of my weakest points, so I have been using AI for writing some of the sections of the report. I used 3 different AI tools, namely Gemini, Claude and ChatGPT. Since my conversation with Gemini was so extremely long, it is impossible to put the entire conversation in a document and attach it to the assignment. Therefore, I asked Gemini to give a high level summary of what kind of things I have been asking to it. In the summary you can read that I have also been using it to integrate some findings about the biological model into the explanations for Part I of the assignment. The reason for this is that Irina Kalmykova did a lot of statistical analysis for Part I. She has a way stronger background in statistics than I have, so I had to catch up with the statistics. Therefore, I also used Gemini to explain the statistics to me, so that I can perform the same analysis on the biological model that I developed for Part II of the assignment. Gemini helped me to integrate this analysis into the already existing analysis for Part I so that it aligned with the rest of the explanations. Below you can find what Gemini gave as a summary:

My prompt (sorry it is not the best prompt): Please now summarise what I have been asking you in this conversation. Mention what kind of things you helped me with. You do not have to mention every single detail, but provide an overview of what you helped me with so that I can put that in the AI disclaimer for my assignment.

The answer from Gemini:

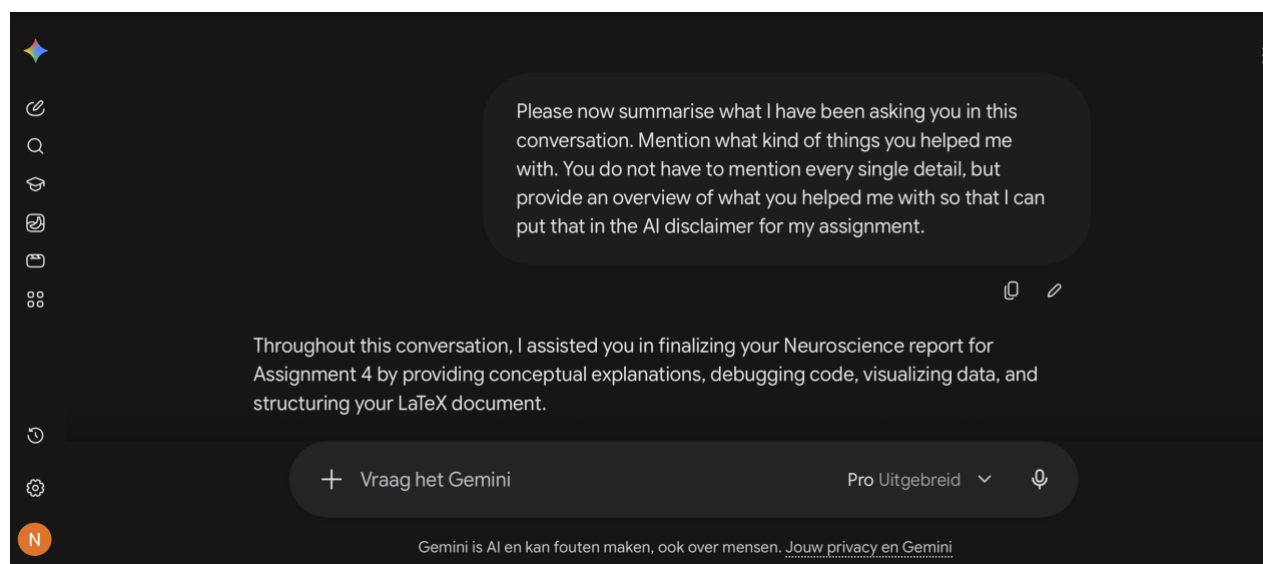
Throughout this conversation, I assisted you in finalizing your Neuroscience report for Assignment 4 by providing conceptual explanations, debugging code, visualizing data, and structuring your LaTeX document.

Here is a concise breakdown of what we worked on together that you can adapt for your AI disclaimer:

- **Model Architecture & Coding:** I clarified the biological and mathematical reasoning for using asymmetric frequency pooling (pool_freq) in the higher auditory layers to preserve temporal fidelity for 1.0-second audio snippets. I also helped resolve a minor command-line interface error for your training script.
- **Data Visualization & Dataframes:** I provided Python and Matplotlib code to accurately plot the branching architecture of your biological model, ensuring the parallel

LBelt and MBelt streams were represented at the same structural depth in both the figures and the summary pandas DataFrames.

- **Statistical Interpretation:** I helped interpret the permutation test results, explaining p -value limits, chance levels, and training deltas. I also analyzed the biological model's unique results, particularly how it broke the expected trend by achieving high STG alignment despite lower linear category decoding (LDA) compared to the inspired model.
 - **Text Integration:** I wove the specific statistical and visual findings for your custom biological model seamlessly into your teammate's existing drafts for the Results section (covering depth slopes, loudness confounds, MDS domain/loudness maps, and LDA projections) while preserving their original logic and tone.
 - **Scientific Writing & LaTeX Formatting:** I transformed your raw notes on anatomical mapping, Capacity Scaling, and Local Response Normalization into cohesive, academic text. I drafted the final LaTeX blocks for your Introduction, Methods, Discussion, Future Work, and Conclusion sections, ensuring they aligned with the course lectures, and formatted your `\bibitem` bibliography entries.
- Good luck with the final submission of your multiscale modeling assignment!



The second AI tool that I used was Claude. I used 2 separate chats with Claude, but they were very short compared to the long chat with Gemini. In these chats Claude helped me with debugging code and explaining me why some parts of the code could possible cause problems later on. Based on that I could look into my code and find the specific problematic places and then adjust them to avoid errors later on. This saved a lot of time and headaches.

Below you can find the prompt that I used to obtain the summaries, and the summaries of the conversations themselves.

My prompt (again not a very good prompt but it worked): Please give me a high level summary of what you helped me with in this chat so that I can put this in the AI disclaimer for my assignment.

The answer from Claude about the first chat:
Here is a draft you can adapt:

AI use disclosure: I used Claude (Anthropic) as a support tool while working on Part II of this assignment. I wrote the BiologicalModel architecture myself. Claude's help was limited to two things:

1. **Debugging a training error.** When train.py failed with an MPS backend error ("only supported on devices with unified memory"), Claude explained that the problem was in the PyTorch/hardware setup, not in my model code. It suggested checking the Python environment's architecture (arm64 vs. x86_64) and running on CPU with --device cpu as a workaround.

2. **Advice on using extract_activations with my model.** Claude explained how the provided function handles layers in my model. Conv and BatchNorm layers both count as trainable, so each block gives two entries, one before and one after the activation, and only one should be kept per block for fair comparison. It also pointed out that the parallel LBelt/MBelt branches need consistent depth handling, and that GRU and conv activations must be flattened or averaged over time before building RDMs. Claude initially misjudged one point about the shared activation module and corrected it within the conversation. I checked the final advice against the provided code. Claude did not write my analysis, figures, or report text. All decisions about methodology and interpretation are my own.

Adjust the wording to match what you actually did. If you later use Claude for other parts, such as the RSA code or the report, add those to the disclosure too.

MPS backend unified memory error

2 days ago

Please give me a high level summary of what you helped me with in this chat so that I can put this in the AI disclaimer for my assignment.

Drafting a disclaimer noting an earlier overstatement.

Here is a draft you can adapt:

AI use disclosure: I used Claude (Anthropic) as a support tool while working on Part II of this assignment. I wrote the `BiologicalModel` architecture myself. Claude's help was limited to two things:

Write a message...

Claude is AI and can make mistakes. Please double-check responses.

Sonnet 5.5 Medium

The answer from Claude about the second chat: Here is a draft you can adapt. Edit it so it matches what you actually did and used.

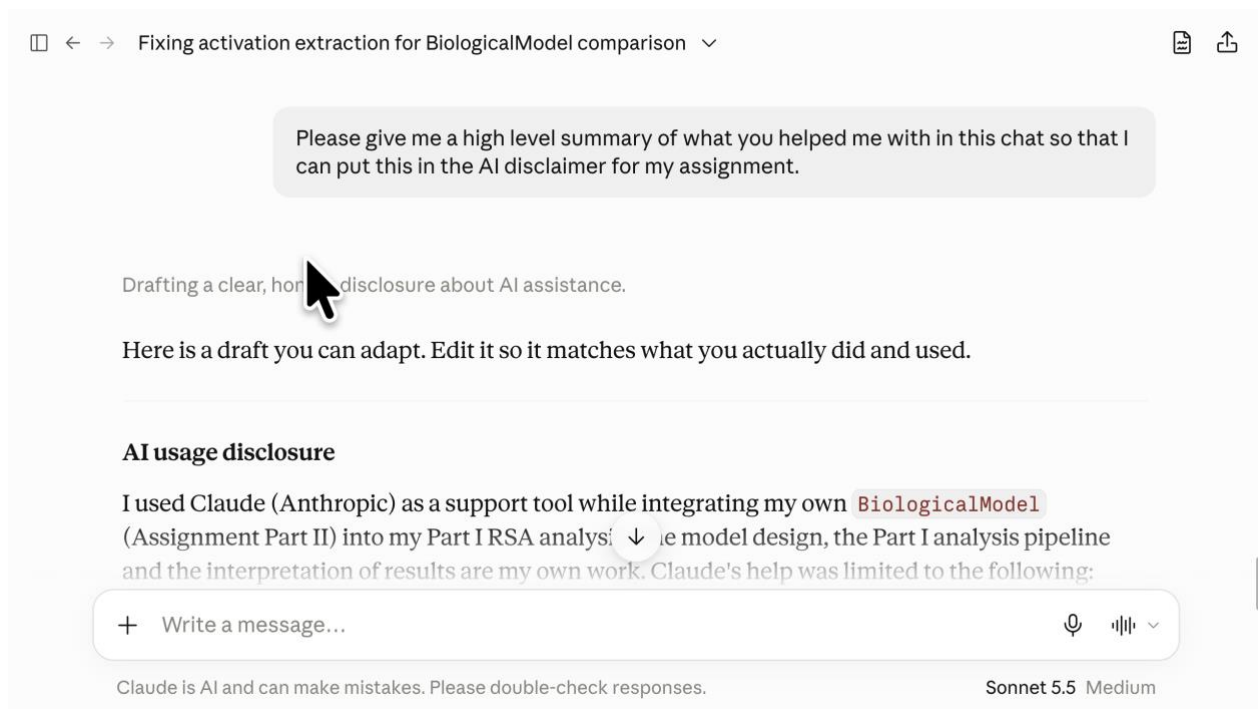
AI usage disclosure

I used Claude (Anthropic) as a support tool while integrating my own BiologicalModel (Assignment Part II) into my Part I RSA analysis. The model design, the Part I analysis pipeline and the interpretation of results are my own work. Claude's help was limited to the following:

- **Code review and debugging guidance.** Claude read my models.py, core.py, train.py and my Part I notebook. It explained how `extract_activations` behaves for my BiologicalModel. Advice from an earlier conversation had assumed BatchNorm layers, but my model uses LocalResponseNorm, so Claude corrected it. The result is that no layer filtering is needed and all seven layers (`conv_a1`, `conv_lbelt`, `conv_mbelt`, `conv_pbelt`, `conv_a4`, `rnn`, `classifier`) are returned as post-activation outputs.
- **Workflow advice.** Claude explained that the model needs to be retrained with train.py (using `--models biological` and the same number of runs and settings as the provided models) so that the checkpoints match the current architecture.
- **Notebook integration plan.** Claude listed which notebook cells need "biological" added to the model lists, plots and statistics. It also pointed out that the multiple-comparison corrections change with four models, and that the interpretation text must be rewritten after rerunning.
- **Depth-analysis suggestion.** Claude proposed a small helper function that gives the two parallel belt branches the same relative depth, instead of treating them as consecutive layers.
- **Report guidance.** Claude suggested what to describe in the Part II section of the report.

Claude did not run my notebook or produce my results. All results, statistical interpretation and written text are mine, and I checked and adapted the suggested code before using it.

Adjust the last paragraph if you did anything differently, for example if Claude also helped with other parts of the notebook in another conversation.



Finally, I used ChatGPT for creating the first draft structure for the report. We initially have been writing all our explanations and comments about the code in one single big notebook file. However, at some point we realised that we also have to put all these explanations in a report in a nice way. Putting it all together in a nice way and extracting all the explanations from all the different markdown cells in the report was extremely difficult and time consuming. Therefore, we asked ChatGPT to extract the explanations from the notebook and create a draft report with all these explanations. However, we deliberately asked to not change the content of the explanations and not to write the sections for us that were not covered yet by explanations from the notebook. In this way, we tried to make sure that AI was not writing our report, just giving a first structure, so that we could built on top of that and use our own explanations. This helped us a lot, while still ensuring that our report is really our own report (although it is extremely long, my apologies for that!).

This was my conversation with ChatGPT:

My prompt: In the file 'assignment4_part1.ipynb' you can find the code for Part I (incorporating the analysis for Part II) of the assignment, as well as all the explanations. I will add the explanations about how I created the biological model in Part II later. But I need to put literally all the explanations and figures from the notebook in our report. Please follow the guidelines from the assignment regarding the report and while doing that, create a draft for our report. Incorporate all the explanations that are in the notebook, and put placeholders for the sections that still need to be written by my teammate and me. You do not have to write complete new sections for us, just make sure that the report is nicely structured and includes ALL our figures, and explanations from the notebook.

The answer from ChatGPT: Done. I created a **Word draft of the report** based directly on assignment4_part1.ipynb and the assignment guidelines.

The draft:

- follows the required report structure: **Introduction** → **Methods** → **Results** → **Discussion** → **Future Work** → **Conclusion**
- incorporates the notebook's **explanations and interpretations**
- includes **all five figures generated in the notebook**
- includes the major **results tables**, including training effects, permutation tests, architecture comparisons, depth analysis, and loudness analysis
- preserves the notebook's numerical results and statistical interpretations
- incorporates the assignment's requirements concerning RSA, confidence intervals, multiple-comparison corrections, training effects, architecture, depth, and MDS/t-SNE-style visualization
- includes clearly marked **[PLACEHOLDER – TO BE COMPLETED]** sections for the biological-model construction, study-type justification, additional discussion, limitations, future work, conclusion, and references
- keeps the Part II material open so you and your teammate can add the explanation of how the biological model was actually designed.

Report draft

Download the complete report draft (.docx)

I deliberately **did not invent the missing Part II details**. The places where your own biological-model explanation or additional interpretation is needed are explicitly marked, so you can add those later without having to restructure the report.

My follow up prompt: Can you turn this entire draft report into LaTeX code for OverLeaf?

The answer from ChatGPT: Absolutely. I converted the **entire Word draft** into an Overleaf-ready LaTeX project, including:

- all report sections and subsections
- all explanations from the draft
- all tables
- all **5 figures**
- all Part II placeholders
- the discussion/future-work/conclusion placeholders
- a `\placeholder{...}` command that makes unfinished sections clearly visible in Overleaf
- longtable formatting for the larger result tables so they can span multiple pages

Download the Overleaf project

Download the complete Overleaf project (.zip)

The ZIP contains:

main.tex

figure_cell11_output1.png

figure_cell35_output0.png

figure_cell45_output0.png

figure_cell47_output0.png

figure_cell50_output0.png

How to use it in Overleaf

1. Download the ZIP above.
2. Go to Overleaf and create **New Project** → **Upload Project**.
3. Upload assignment4_overleaf_project.zip.
4. Overleaf should automatically detect main.tex as the main document.
5. Compile with **pdfLaTeX**.

I also kept the source as a normal, editable LaTeX document rather than converting the Word document into a visually fixed PDF-style layout, so you can freely modify the text, tables, captions, references, and placeholders.

This follow up question was really useful to get the draft report into a good LaTeX format. In previous projects I have been struggling a lot with LaTeX code, so fortunately this worked immediately so I could focus on the more important parts of this assignment, namely writing the report itself and doing the analysis, and creating our own biologically inspired model.

Finally, when all the work was done, I asked Claude to generate a README.md file for our assignment. I usually prefer not to generate it with AI, but we had quite a lot of time pressure since we had only 6 days for this assignment and we really liked the assignment so we wanted to make it as good as possible. Therefore, this time I used Claude to generate a basis for our README file, after which I adjusted a lot of things myself. This is the prompt that I gave to Claude:

Please generate a readme.md file for the repository of my assignment. These are the minimal requirements for the readme file:

- Describe the objective of the assignment
- Provide instructions on how to run your code
- Include any other relevant notes or dependencies

I have added a lot of the important files so that you can see how everything works. In same folder, there are also folders named *pycache*, *data* (the folder with the ESC50 dataset, *santoro_sounds*, *yamnet_embeddings*, *stg_betas_test.mat*, *stg_betas_train.mat*, *test_sound_keys.mat* and *train_sound_keys.mat*), *img*, *models* (with all the trained models), *practical* (the code that we created during the practical, which was a starting point for our assignment), *results* (files with results of our experiments) and a folder called *Use_of_LLM_disclosures* which contains disclaimers/disclosures about how we used AI/LLM's. In the readme, also please list everything that is in the repository and give a description of what everything is and what the dependencies are. Make the readme file as complete as possible.

Claude gave me a first version of a README.md file. However, it was a bit too complete/extensive, so I deleted some things. However, I also really like being very complete so I also kept a lot of things. But I also had to change some things because not everything was correct.

I hope that this disclosure/disclaimer about the use of LLM's is complete enough. I apologise for making a bit more use of AI than I would usually do. But I really enjoyed the assignment and learned a lot from it, and I have been spending a couple of days on

it, including working during the night, so although AI helped a lot, my teammates and I have been doing a lot of work ourselves.